

Program 8

Deploy a Node.js App in Kubernetes Using ConfigMap & Pod

Project Structure

```
|— server.js
|— nodejs-configmap.yaml
|— nodejs-pod.yaml
```

Step 1: Create server.js File

Step 2: Create nodejs-configmap.yaml

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~$ cd DevOps/
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ ls
Dockerfile package.json package-lock.json program1 program2 program3 program4 program5 project-6 project-7 project-8
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ cd project-8
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ ls
nodejs-configmap.yaml nodejs-pod.yaml server.js
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ nano server.js
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ nano nodejs-configmap.yaml
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ nano nodejs-pods.yaml
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ nano nodejs-pod.yaml
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ rm nodejs-pods.yaml
rm: cannot remove 'nodejs-pods.yaml': No such file or directory
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ ls
nodejs-configmap.yaml nodejs-pod.yaml server.js
```

Step 3: Create nodejs-pod.yaml

```

1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-62zkk  1/1     Running   0           10m
nginx-deployment-6f9664446b-jfsmg   1/1     Running   0           10m
program-8                           0/1     ContainerCreating   0           7s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-62zkk  1/1     Running   0           10m
nginx-deployment-6f9664446b-jfsmg   1/1     Running   0           10m
program-8                           0/1     ContainerCreating   0           14s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-62zkk  1/1     Running   0           10m
nginx-deployment-6f9664446b-jfsmg   1/1     Running   0           10m
program-8                           0/1     ContainerCreating   0           27s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-62zkk  1/1     Running   0           11m
nginx-deployment-6f9664446b-jfsmg   1/1     Running   0           11m
program-8                           0/1     ContainerCreating   0           38s
^C1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-62zkk  1/1     Running   0           11m
nginx-deployment-6f9664446b-jfsmg   1/1     Running   0           11m

```

Step 4: Apply YAML Files

kubectl delete pod program-8

kubectl apply -f nodejs-pod.yaml

```

program-8                           0/1     ContainerCreating   0           60s
^C1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ ^C
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-62zkk  1/1     Running   0           11m
nginx-deployment-6f9664446b-jfsmg   1/1     Running   0           11m
program-8                           1/1     Running   0           84s
^C1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-62zkk  1/1     Running   0           11m
nginx-deployment-6f9664446b-jfsmg   1/1     Running   0           11m
program-8                           1/1     Running   0           92s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
^C1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
^C1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000

```

Step 5: Access the Node.js App

Port forward:

kubectl port-forward pod/program-8 3000:3000

